

CPU-Scheduling

Definition (Please input your data here.)

TIME QUANTUM

2

Scenario D: Interactive-style fairness



Load

Process name	Arrival	CPU burst	
 P1	0	8	
 P2	0	8	
 P3	0	8	
 P4	0	8	

+ ADD A NEW PROCESS

RUN SIMULATION



Detailed Results (Round Robin)

Process Name	Arrival	Burst	WT (RR)	TAT (RR)	RT (RR)
P1	0	8	18	26	0
P2	0	8	20	28	2
P3	0	8	22	30	4
P4	0	8	24	32	6
AVERAGES:			21.00	29.00	3.00



Process Name	Arrival	Burst	WT (SRTF)	TAT (SRTF)	RT (SRTF)
P1	0	8	0	8	0
P2	0	8	8	16	8
P3	0	8	16	24	16
P4	0	8	24	32	24
AVERAGES:			12.00	20.00	12.00

Performance Comparison Summary

AVG WAITING TIME

21.00 vs **12.00**

AVG TURNAROUND TIME

29.00 vs **20.00**

AVG RESPONSE TIME

3.00 vs **12.00**

Final Conclusion & Analytical Comparison

Fairness versus efficiency

Round Robin ensures high fairness by sharing CPU time equally through time slicing, preventing starvation. However, frequent context switching can reduce overall efficiency. **SRTF** maximizes efficiency by prioritizing short jobs. In this run, **SRTF** proved more efficient with an average waiting time of **12.00** compared to **RR**'s **21.00**.

Effect of time slicing versus shortest-job preference

Time slicing (RR) guarantees regular CPU allocation, making it ideal for interactive systems. **Shortest-job preference (SRTF)** heavily optimizes for throughput by quickly finishing tasks. The simulation reflects this with RR yielding a turnaround time of **29.00**, while SRTF achieved **20.00**.

Effect on first response time

Round Robin generally provides a better and more predictable first response time. **SRTF** provides lightning-fast response times for short jobs. Here, **RR** provided a better average response time (**3.00**) than **SRTF** (**12.00**), proving its strength for interactive responsiveness.

Effect of quantum size on Round Robin behavior

A smaller quantum improves response time but increases context switching overhead, hurting overall throughput. A larger quantum reduces overhead but makes RR degrade into First-Come-First-Served (FCFS). The chosen small quantum of **2** heavily favors responsiveness but causes frequent context switches.

Whether SRTF gives a strong advantage to short jobs

Yes, SRTF heavily favors short jobs by immediately preempting currently running longer jobs. This simulation confirms the theory: SRTF achieved a lower average waiting time (**12.00** vs **21.00**), strongly advantaging the shorter processes.

CPU-Scheduling

Definition (Please input your data here.)

TIME QUANTUM

3

Scenario C: Short-job-heavy case



Load

Process name	Arrival	CPU burst	
 P1	0	15	
 P2	1	2	
 P3	2	1	
 P4	3	3	
 P5	4	2	

+ ADD A NEW PROCESS

RUN SIMULATION

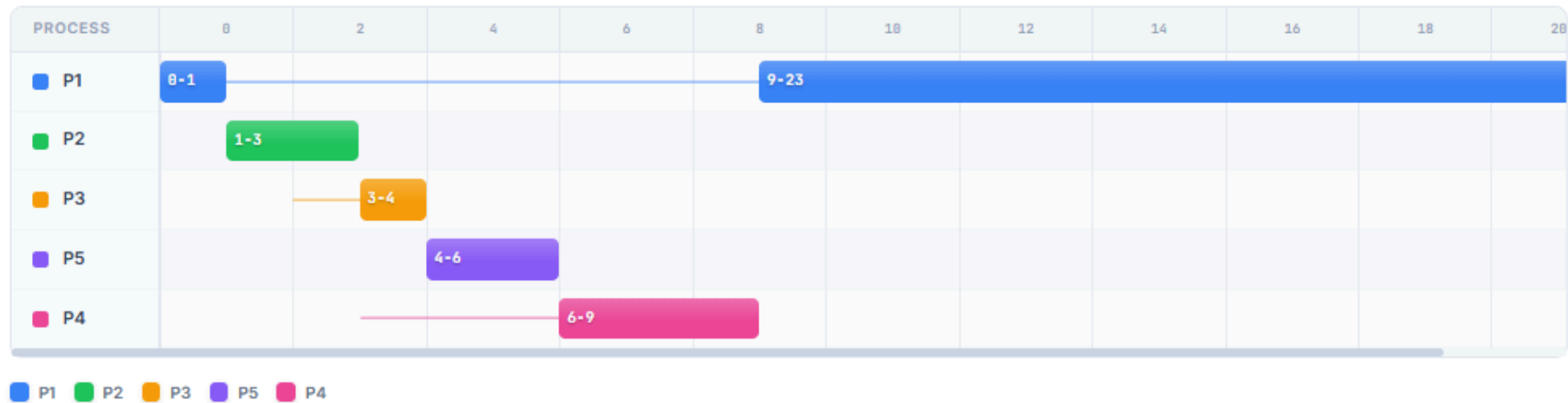
SRTF COMPARISON

⏮ ⏪ ⏩ ⏭ 🔍 1x ⏭ ⏮ ⏲ t=23/23



Process Name	Arrival	Burst	WT (RR)	TAT (RR)	RT (RR)
P1	0	15	8	23	0
P2	1	2	2	4	2
P3	2	1	3	4	3
P4	3	3	3	6	3
P5	4	2	8	10	8
AVERAGES:			4.80	9.40	3.20

SRTF - GANTT CHART



Detailed Results (SRTF)

Process Name	Arrival	Burst	WT (SRTF)	TAT (SRTF)	RT (SRTF)
P1	0	15	8	23	0
P2	1	2	0	2	0
P3	2	1	1	2	1
P4	3	3	3	6	3
P5	4	2	0	2	0
AVERAGES:			2.40	7.00	0.80

Performance Comparison Summary

AVG WAITING TIME

4.80 vs **2.40**

AVG TURNAROUND TIME

9.40 vs **7.00**

AVG RESPONSE TIME

3.20 vs **0.80**

Final Conclusion & Analytical Comparison

Fairness versus efficiency

Round Robin ensures high fairness by sharing CPU time equally through time slicing, preventing starvation. However, frequent context switching can reduce overall efficiency. **SRTF** maximizes efficiency by prioritizing short jobs. In this run, **SRTF** proved more efficient with an average waiting time of **2.40** compared to **RR**'s **4.80**.

Effect of time slicing versus shortest-job preference

Time slicing (RR) guarantees regular CPU allocation, making it ideal for interactive systems. **Shortest-job preference (SRTF)** heavily optimizes for throughput by quickly finishing tasks. The simulation reflects this with RR yielding a turnaround time of **9.40**, while SRTF achieved **7.00**.

Effect on first response time

Round Robin generally provides a better and more predictable first response time. **SRTF** provides lightning-fast response times for short jobs. Here, **SRTF** provided a better average response time (**0.80**) than **RR** (**3.20**), likely because short jobs quickly cleared out before long ones.

Effect of quantum size on Round Robin behavior

A smaller quantum improves response time but increases context switching overhead, hurting overall throughput. A larger quantum reduces overhead but makes RR degrade into First-Come-First-Served (FCFS). The chosen small quantum of **3** heavily favors responsiveness but causes frequent context switches.

Whether SRTF gives a strong advantage to short jobs

Yes, SRTF heavily favors short jobs by immediately preempting currently running longer jobs. This simulation confirms the theory: SRTF achieved a lower average waiting time (**2.40** vs **4.80**), strongly advantaging the shorter processes.

CPU-Scheduling

Definition (Please input your data here.)

TIME QUANTUM

1

Scenario B: Quantum sensitivity case ▾

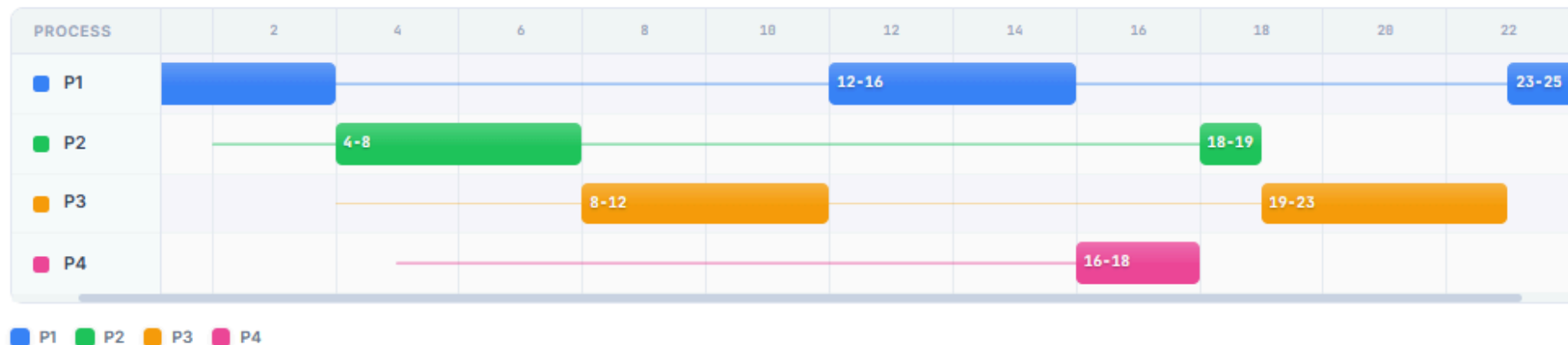
Load

Process name	Arrival	CPU burst	
 P1	0	10	
 P2	2	5	
 P3	4	8	
 P4	5	2	

+ ADD A NEW PROCESS

RUN SIMULATION

⏮ ⏪ ⏩ ⏭ - 5x + t=25/25



Process Name	Arrival	Burst	WT (RR)	TAT (RR)	RT (RR)
P1	0	10	15	25	0
P2	2	5	12	17	2
P3	4	8	11	19	4
P4	5	2	11	13	11
AVERAGES:			12.25	18.50	4.25

Performance Comparison Summary

AVG WAITING TIME

12.25 vs **5.50**

AVG TURNAROUND TIME

18.50 vs **11.75**

AVG RESPONSE TIME

4.25 vs **3.75**

Final Conclusion & Analytical Comparison

Fairness versus efficiency

Round Robin ensures high fairness by sharing CPU time equally through time slicing, preventing starvation. However, frequent context switching can reduce overall efficiency. **SRTF** maximizes efficiency by prioritizing short jobs. In this run, **SRTF** proved more efficient with an average waiting time of **5.50** compared to **RR**'s **12.25**.

Effect of time slicing versus shortest-job preference

Time slicing (RR) guarantees regular CPU allocation, making it ideal for interactive systems. **Shortest-job preference (SRTF)** heavily optimizes for throughput by quickly finishing tasks. The simulation reflects this with RR yielding a turnaround time of **18.50**, while SRTF achieved **11.75**.

Effect on first response time

Round Robin generally provides a better and more predictable first response time. **SRTF** provides lightning-fast response times for short jobs. Here, **SRTF** provided a better average response time (**3.75**) than **RR** (**4.25**), likely because short jobs quickly cleared out before long ones.

Effect of quantum size on Round Robin behavior

A smaller quantum improves response time but increases context switching overhead, hurting overall throughput. A larger quantum reduces overhead but makes RR degrade into First-Come-First-Served (FCFS). The chosen larger quantum of **4** reduces context switches but makes RR behave more like FCFS.

Whether SRTF gives a strong advantage to short jobs

Yes, SRTF heavily favors short jobs by immediately preempting currently running longer jobs. This simulation confirms the theory: SRTF achieved a lower average waiting time (**5.50** vs **12.25**), strongly advantaging the shorter processes.

Performance Comparison Summary

AVG WAITING TIME

12.25 vs 5.50

AVG TURNAROUND TIME

18.50 vs 11.75

AVG RESPONSE TIME

4.25 vs 3.75

Final Conclusion & Analytical Comparison

Fairness versus efficiency

Round Robin ensures high fairness by sharing CPU time equally through time slicing, preventing starvation. However, frequent context switching can reduce overall efficiency. **SRTF** maximizes efficiency by prioritizing short jobs. In this run, **SRTF** proved more efficient with an average waiting time of **5.50** compared to **RR**'s **12.25**.

Effect of time slicing versus shortest-job preference

Time slicing (RR) guarantees regular CPU allocation, making it ideal for interactive systems. **Shortest-job preference (SRTF)** heavily optimizes for throughput by quickly finishing tasks. The simulation reflects this with RR yielding a turnaround time of **18.50**, while SRTF achieved **11.75**.

Effect on first response time

Round Robin generally provides a better and more predictable first response time. **SRTF** provides lightning-fast response times for short jobs. Here, **SRTF** provided a better average response time (**3.75**) than **RR** (**4.25**), likely because short jobs quickly cleared out before long ones.

Effect of quantum size on Round Robin behavior

A smaller quantum improves response time but increases context switching overhead, hurting overall throughput. A larger quantum reduces overhead but makes RR degrade into First-Come-First-Served (FCFS). The chosen larger quantum of **4** reduces context switches but makes RR behave more like FCFS.

Whether SRTF gives a strong advantage to short jobs

Yes, SRTF heavily favors short jobs by immediately preempting currently running longer jobs. This simulation confirms the theory: SRTF achieved a lower average waiting time (**5.50** vs **12.25**), strongly advantaging the shorter processes.

CPU-Scheduling

Definition (Please input your data here.)

TIME QUANTUM

4

Scenario A: Basic mixed workload



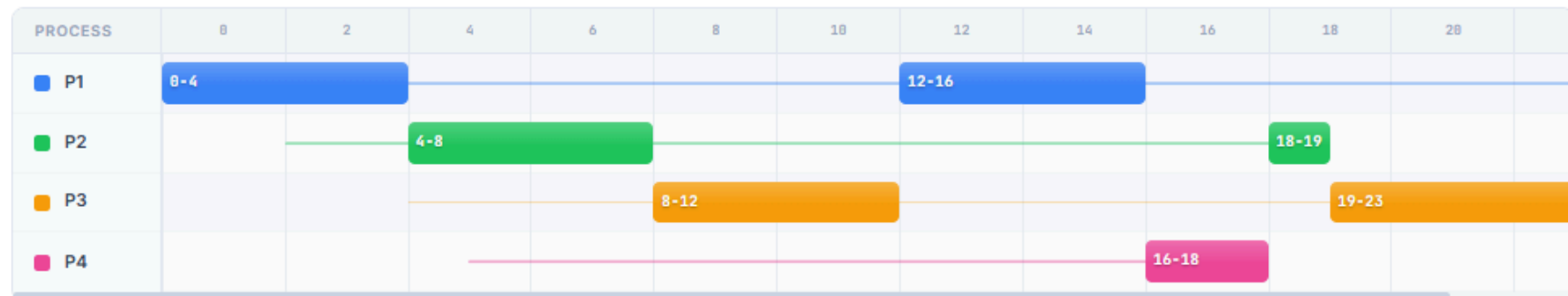
Load

Process name	Arrival	CPU burst	
 P1	0	10	
 P2	2	5	
 P3	4	8	
 P4	5	2	

+ ADD A NEW PROCESS

RUN SIMULATION

ROUND ROBIN - GANTT CHART



■ P1 ■ P2 ■ P3 ■ P4

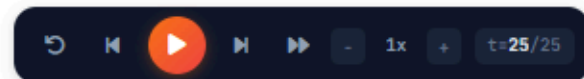
Detailed Results (Round Robin)

Process Name	Arrival	Burst	WT (RR)	TAT (RR)	RT (RR)
P1	0	10	15	25	0
P2	2	5	12	17	2
P3	4	8	11	19	4
P4	5	2	11	13	11
AVERAGES:			12.25	18.50	4.25

ROUND ROBIN

SRTF COMPARISON

SRTF - GANTT CHART



Detailed Results (SRTF)

Process Name	Arrival	Burst	WT (SRTF)	TAT (SRTF)	RT (SRTF)
P1	0	10	7	17	0
P2	2	5	0	5	0
P3	4	8	13	21	13
P4	5	2	2	4	2
AVERAGES:			5.50	11.75	3.75